Part 1: Project Plan

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Project title: Forecasting Weekly Brent Crude Direction with LLM-Derived Oil Supply Risk: Do Chokepoint and Country Level Signals Add Value Beyond Geopolitical Risk Indices?

Project description

The 2026 Strait of Hormuz crisis showed how quickly threats to a single shipping chokepoint can move the global oil price. Yet existing geopolitical risk indices measure risk by country or broad region and oil supply risk at the level of individual producers and chokepoints has not yet been accounted for. This project tests whether such signals improve forecasts of Brent's weekly direction given the current war in the middle east and the importance of the safety of chokepoints. Brent crude is the main international oil price benchmark and sharp moves in it feed through to inflation, company costs and trading decisions. Oil prices are hard to forecast: a simple no change forecast is difficult to beat (Alquist, Kilian and Vigfusson, 2013) and many large price rises have come from fear of future supply shortfalls rather than from barrels actually lost (Kilian, 2009). Recent work measures geopolitical risk from news with large language models (LLMs) most notably the AI-GPR index (Iacoviello and Tong, 2026). These indices, however, measure general risk by country or broad region and were not built or tested as forecasting inputs.

This project builds a machine learning system that predicts the direction of Brent over the next five trading days: buy (return above +1.5%), sell (below −1.5%) or hold. It uses an ensemble of Random Forest, XGBoost and LightGBM, evaluated by walk-forward backtesting. Three nested models are compared. **Model A** uses market fundamentals only (price and technical indicators, supply, macro, demand and positioning data). **Model B** adds the full AI-GPR suite. Model C adds oil supply risk scores for individual producer countries and shipping chokepoints (Hormuz, Bab-el-Mandeb, Suez), generated by an LLM from GDELT news, together with war-risk insurance listings. The comparison of **Model B** with **Model A** shows whether existing geopolitical risk measures help at all but the main result is the comparison of **Model C with Model B** which shows whether the finer-grained signals add value beyond the best existing index. The research question is: do oil-supply-specific, country and chokepoint-level features improve weekly Brent direction forecasts beyond a model that already uses the full AI-GPR suite? Differences are tested with the Diebold–Mariano test, explained with SHAP values and examined in a case study of the 2026 Strait of Hormuz crisis which falls after the LLM's training cutoff.

Deliverables

Basic

- **B1.** A weekly dataset of Model A features from verified public sources (EIA, FRED, CFTC, Baker Hughes, OPEC production) with each feature lagged to its publication date to prevent look ahead.
- **B2.** Model A: Random Forest, XGBoost, LightGBM and their ensemble, evaluated by walk-forward backtesting against three baselines: always-hold (no-change), majority class, and last week's class.).
- **B3.** An evaluation framework: accuracy, macro-F1, confusion matrices and Diebold–Mariano tests with a small-sample correction.

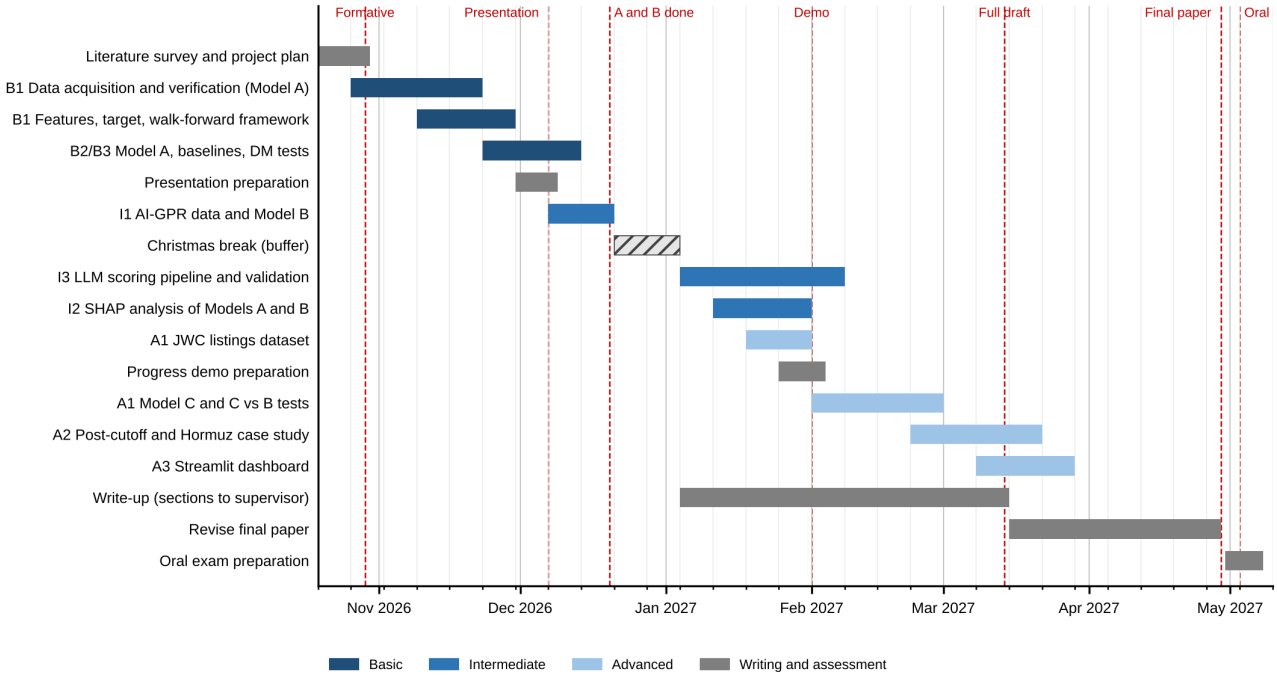
Intermediate

- **I1.** Model B: Model A plus the full published AI-GPR suite (headline index, threats and acts, Oil GPR with its regional series, and AI-GPR's country series), with B versus A tests. No new risk scores are added at this stage.
- **I2.** SHAP analysis of Models A and B: global importance, importance over time, and explanations of individual weeks.
- **I3.** High-risk evaluation of Models A and B: regression of the loss difference on a high-risk indicator, and sign tests in crisis periods, to test whether geopolitical information helps mainly in high-risk weeks.

Advanced

- **A1.** Model C: an LLM scoring pipeline (GDELT news filtered to oil producers and chokepoints, scored by GPT-4o mini with a date-anchored prompt and checked against a hand-labelled sample), producing producer and chokepoint scores that are added to Model B with a JWC listed-area flag. C is tested against B overall and in high-risk weeks.
- **A2.** A post-cutoff case study: performance and SHAP explanations for weeks after the LLM's training cutoff, centred on the 2026 Hormuz crisis.
- **A3.** A Streamlit dashboard showing the weekly signal and the features driving it.

Timeline



Models A and B are scheduled to be finished before the Christmas break. The presentation (week of 7 December) can then report Model A results and the progress demo (week of 1 February) can show Models A and B alongside a working scoring pipeline. Model C, the case study and the dashboard follow in February and March, leaving time for a full draft before the final deadline of 29 April.

Preliminary preparation

- **Data access.** Register for EIA and FRED API keys; confirm OPEC production via EIA international data; script the Baker Hughes and CFTC downloads; choose a demand proxy (a free PMI series or Kilian's global real activity index); check whether Brent futures curve data is available for the futures-basis feature.
- **Publication lags.** Build a table of release dates for every source, so each feature only uses data available on the forecast date.
- **AI-GPR.** Download the series from the authors' companion site and confirm which regional and country series are public and how often they are updated.
- **LLM pipeline.** Learn GDELT access through Google BigQuery, estimate GPT-4o mini API costs before scaling up, and design a date-anchored prompt.
- **JWC listings.** Find an archive of past Listed Areas circulars to build a dated history.
- **Methods.** HAC standard errors for the Diebold–Mariano test (statsmodels), TreeSHAP in the shap library, and handling class imbalance in the hold class.

Part 2: Literature Survey

1. Introduction and key terms

This survey covers three themes: what drives oil prices and how far they can be forecast; how geopolitical risk can be measured from news text and the machine learning and evaluation methods needed to test whether such measures add predictive value. The final section assesses these themes and sets out how they shape the project. The main terms are defined first.

Brent crude and directional forecasting. Brent is the main international crude oil benchmark. Rather than predicting the price level, this project forecasts the direction of the five-trading-day return in three classes: buy (above +1.5%), sell (below −1.5%) and hold. Direction is a recognised way of judging oil forecasts. Alquist, Kilian and Vigfusson (2013) report the success ratio, the share of correctly predicted price changes, alongside the mean squared prediction error and test it with the sign test of Pesaran and Timmermann (2009).

Oil price shocks. Kilian (2009) separates three kinds of shock: oil supply shocks (changes in the physical availability of crude oil), aggregate demand shocks (the global business cycle raising demand for all industrial commodities) and oil-specific demand shocks which are designed to capture precautionary demand driven by concerns about future supply shortfalls.

Geopolitical risk. Caldara and Iacoviello (2022, p. 1197) define it as "the threat, realization and escalation of adverse events associated with wars, terrorism, and any tensions among states and political actors that affect the peaceful course of international relations." **This project uses the narrower term oil supply risk for the risk that such an event disrupts the production or transport of crude at a specific producer or chokepoint.**

Chokepoints and war risk listings. Chokepoints are narrow sea passages that carry a large share of seaborne oil, such as the Strait of Hormuz, Bab-el-Mandeb and the Suez Canal [EIA, *World Oil Transit Chokepoints*, year]. Because tankers must pass through them they are exposed to attack, seizure or blockade, as in the Iran–Iraq "Tanker War" of the 1980s (Navias, 2019) the attacks on shipping in the Red Sea from late 2023 (UNCTAD, 2024). The Joint War Committee (JWC) of the London marine insurance market publishes Listed Areas. Vessels entering them must notify their insurers and may be charged additional war risk premiums [JWC, Listed Areas circular, number and date]. A new listing therefore signals that risk to shipping has become costly to insure, not just reported.

LLMs as text scorers. Pretrained large language models can classify or score text from instructions alone, without task-specific training (Lopez-Lira and Tang, 2023). The model's training cutoff matters: a model scoring news published before its cutoff may already know how events ended, a look ahead bias. Scoring news published after the cutoff removes this bias but not the distraction effect where the model's general knowledge of a named entity colours its score (Glasserman and Lin, 2023).

Walk forward evaluation. Each model is trained only on data before a test period which then rolls forward through time, so no future information enters training. This matters because shuffled cross-validation gives over optimistic estimates on time series, and for non-stationary series, methods that keep time order and test over several periods give the most accurate estimates (Cerqueira, Torgo and Mozetič, 2020).

2. Theme 1: What drives oil prices and can they be forecast?

Kilian (2009) estimates a structural vector autoregression (VAR) on monthly data from 1973 to 2007, using world oil production, an index of global real activity built from dry cargo shipping rates, and the real oil price. Supply shocks have small and short-lived effects on the price. The large rises

came mostly from global demand, which moves slowly and from precautionary demand which produces sudden spikes such as those of 1979 and 1990. Baumeister and Kilian (2016) review forty years of oil price swings and reach a similar conclusion: most major moves came from demand, including precautionary inventory demand and geopolitical events mattered mainly through expectations. They note that the Iran–Iraq War "had little effect on the price of oil in the 1980s, notwithstanding considerable damage to oil shipping in the Persian Gulf with as many as 30 attacks on oil tankers in a given month", and that inventory demand rises "only when geopolitical turmoil coincides with expectations of strong demand for crude oil and tight oil supplies". They also argue that even when the drivers of oil prices are understood, predicting those drivers is very difficult, so wars and recessions will keep producing surprises.

On forecasting, Alquist, Kilian and Vigfusson (2013) compare futures prices, surveys, time series models and VARs using recursive out-of-sample tests. The no-change forecast proves hard to beat, and futures and expert surveys do not consistently improve on it. The gains they do find are at short horizons, from VARs that include inventories and global activity and from recent changes in non-oil commodity prices. Both this literature and Kilian (2009) work with monthly data so evidence at weekly horizons is limited.

Three points follow for this project. First, any model has to beat simple benchmarks: the no-change forecast (always predicting hold), the most frequent class, and last week's class. In the nested design, Model A is the main benchmark against which Models B and C are judged. Second, the literature leaves room for forward-looking risk measures. Physical supply changes predict little, but precautionary demand responds to expected disruptions, which news may reveal before production data does. This is the channel through which the supply-risk scores in Model C are expected to work. Third, the same literature is a challenge: supply-related events usually matter less than demand, and attacks on shipping alone have not moved prices when supply was plentiful. Model A therefore includes demand-side variables (global activity, inventories and the dollar) so that any gain in Model C is measured against a fair baseline, and the tree-based models can capture whether supply risk matters more when the market is tight.

3. Theme 2: Measuring geopolitical risk from text

Caldara and Iacoviello (2022) construct the Geopolitical Risk (GPR) index by counting each month the share of newspaper articles that discuss adverse geopolitical events and associated threats. The recent index uses ten leading newspapers and starts in 1985 and a historical version from 1900 uses three. In their VAR evidence (Section III.A), higher geopolitical risk is followed by lower investment and employment and by falls in both stock prices and oil prices. This matters here: general geopolitical risk can behave like a demand-side shock for oil, so a single index may combine effects that push oil prices in opposite directions.

Iacoviello and Tong (2026) replace keyword counting with GPT-4o mini which scores around 4.6 million articles from the New York Times, Washington Post and Chicago Tribune since 1960 for the severity of geopolitical risk. Around six major events, the resulting AI-GPR index is less volatile from one day to the next than the keyword index (Section 3). Further classification layers split it into threats and acts (Section 5.1), build an Oil GPR index of geopolitical oil supply disruptions by region, mainly the Middle East and Russia (Section 5.2) and give country and bilateral indices with articles also labelled by event type such as military conflict and sanctions (Section 5.3). A geopolitical oil supply shock identified from the Oil GPR threats index raises oil prices for two months before they fall back gradually (Section 6.2) whereas a general GPR shock is followed by lower oil prices in Caldara and Iacoviello, although the two papers identify shocks differently. The authors also test for hindsight bias by rescoring articles around four major events with a prompt that fixes each article's publication date as the present and find the results nearly indistinguishable from the baseline (Section 4.3). They note that this does not show the model lacks knowledge of what followed only that invoking or denying it barely moves the index.

LLM scores have also been used directly for prediction. Lopez-Lira and Tang (2023) show that GPT-4 scores of news headlines published after the model's knowledge cutoff predict stock market reactions. Glasserman and Lin (2023) warn that backtests of LLM sentiment can be biased when the test period overlaps the model's training data, through look ahead bias and through a "distraction effect" from the model's general knowledge about a company which anonymising headlines reduced.

Dai et al. (2026) are closest in design. They score 8,639 energy news articles from 2020 to 2025 with GPT-4o, Llama 3.2-3b and FinBERT on five sentiment dimensions, aggregate them weekly and use LightGBM to predict the direction of the next week's WTI return, reporting cross-validated AUCs of 0.57 to 0.65. However, their features are general sentiment rather than supply-specific, there is no fundamentals or geopolitical-index baseline, no significance tests are reported and the overlap between the models' training data and their sample is not discussed.

A second related study is Cao et al. (2026)who use GPT-4 to extract entity stances, event types with severity, causal links and sentiment from 4,487 Reuters and Oilprice.com articles. They combine these with a temporal fusion transformer and a graph neural network to forecast the daily Brent price from January 2021 to July 2024 and report a root mean squared error of $5.67 against $6.89 for the transformer without news, a 17.7% improvement with larger gains in event intensive periods. However, their evaluation uses a single chronological train, validation and test split on one asset, the target is the price level rather than direction, the news is filtered on keywords linked to Russia, Ukraine, oil, sanctions and energy and the extracted features are general stances, events and sentiment rather than a supply risk score for particular producers or chokepoints. AI-GPR has a similar limit, since its Oil GPR index divides disruption risk only by broad region (Iacoviello and Tong, 2026). As a news source, GDELT (Leetaru and Schrodt, 2013) offers a global, machine-coded record of events reported in the news from 1979 onwards, drawn from a much wider set of outlets than the three US newspapers behind AI-GPR.

4. Theme 3: Learning and evaluation methods

Tree ensembles. Random forests (Breiman, 2001) average many decision trees each grown on a bootstrap sample using a random subset of features at each split, which lowers the correlation between trees. Gradient boosting (Friedman, 2001) instead adds trees in sequence each fitted to the negative gradient of the loss at the current model (the residuals, for squared error). XGBoost (Chen and Guestrin, 2016) adds a regularised objective that penalises the number of leaves and the size of leaf weights which "helps to smooth the final learnt weights to avoid over-fitting" (Section 2.1), along with shrinkage and column subsampling (Section 2.3). It also learns a default direction for missing values (Section 3.4) which helps when series start at different dates. LightGBM (Ke et al., 2017) speeds up boosting through gradient-based one-side sampling and exclusive feature bundling. In a controlled benchmark on medium-sized tabular datasets Tree-based models remain state-of-the-art on medium-sized tabular data (Grinsztajn, Oyallon and Varoquaux, 2022) although their benchmark excludes time series and datasets with fewer than 3,000 samples, so it supports trees as a strong default here rather than proving they suit this sample size. Breiman (2001) shows that forest error depends on the strength of the individual trees and the correlation between them and the same logic applies to combining models: gains depend on their errors differing. Random forests and boosting build trees differently but XGBoost and LightGBM are both boosting, so the correlation between their errors is reported. Chen and Guestrin (2016, Section 1) also report that competition winners found ensembles beat a well-configured XGBoost "by only a small amount", so single models are reported alongside the ensemble.

Evaluation design. Cerqueira, Torgo and Mozetič (2020) compare eleven ways of estimating forecasting performance, using 174 real time series and three synthetic ones. Blocked cross-validation worked for stationary series, but for non-stationary series the most accurate estimates came from out-of-sample testing repeated over several periods. They also note that shuffling data before cross-validation assumes independent observations and may give overly optimistic estimates. The relationship between oil-market features and returns shifts over time which supports walk-forward testing over many periods rather than a single split or shuffled cross-validation.

Comparing forecasts. Diebold and Mariano (1995) test whether two forecasts are equally accurate using the difference in their losses each period. The loss need not be quadratic, symmetric or continuous and forecast errors may be serially correlated and correlated with each other. This suits a classifier where the loss can be misclassification or a trading cost, and forecasts from closely related models. Their simulations show the test is oversized in small samples, and Harvey, Leybourne and Newbold (1997) propose a correction, which is used here. Because Model A is nested in B, B in C and all are re-estimated in each fold, results are read as comparisons of forecasting methods rather than of underlying models, following Giacomini and White (2006) whose framework requires an estimation window of fixed length, so a rolling window is used. Diebold and Mariano (1995, Section 5) also suggest regressing the loss difference on other variables. Regressing it on a high-risk indicator tests whether any gain is concentrated in crisis weeks which Giacomini and White formalise as a test of conditional predictive ability.

Explainability. Lundberg and Lee (2017) show that Shapley values are the only additive feature attributions that satisfy local accuracy, missingness and consistency and call them SHAP values. Lundberg et al. (2020) give TreeSHAP, which computes them exactly in polynomial time for tree ensembles. This allows both overall feature importance and explanations of single weeks, such as signals during the Hormuz crisis. These show which features drive the model's predictions, not what causes oil prices to move.

5. Assessment and use in this project

Three gaps stand out. First, oil-supply risk is not measured at the level where disruptions happen. The AI-GPR oil index counts blockades of shipping lanes and places Saudi Arabia, Iran and Iraq in a single Middle East series. Its country indices capture any conflict or sanctions, whether or not oil supply is threatened, and no series separates chokepoint risk. Second, text-based risk measures have mostly been assessed through VARs and return regressions rather than as forecasting inputs under walk-forward testing with significance tests. Of the two oil forecasting studies reviewed, Cao et al. (2026) rely on a single split and Dai et al. (2026) report no significance tests and no fundamentals or geopolitical-index baseline. Third, market signals of shipping risk such as war-risk listings do not appear in the forecasting studies reviewed. Analysts already watch chokepoints and insurance markets informally. The contribution is to measure this risk systematically from news and test whether it adds forecasting value.

The project design follows directly. Model A uses the fundamentals that Kilian (2009) and Alquist, Kilian and Vigfusson (2013) identify as informative, including inventories and global activity. Model B adds the full AI-GPR suite, so that Model C is compared with the strongest existing measure. Because AI-GPR is built with the same LLM, differences between B and C cannot come from the choice of model, though they may reflect prompts and aggregation as well as granularity and news source. Model C adds producer and chokepoint scores from GDELT, plus the JWC flag, with a date-anchored prompt like the one Iacoviello and Tong (2026) use in their hindsight test. Evaluation uses rolling walk-forward testing, simple benchmarks (always hold, the most frequent class and last week's class), Diebold–Mariano tests with the Harvey, Leybourne and Newbold (1997) correction, a regression of the loss difference on a high-risk indicator, and SHAP explanations.

Hindsight bias is handled mainly through timing. GPT-4o mini's knowledge extends to October 2023 (OpenAI, 2024), so news from November 2023 onwards, including the Red Sea shipping attacks and the 2026 Hormuz crisis, is scored without knowledge of outcomes. Results for this period give the cleanest test, while earlier results may carry some look-ahead bias (Glasserman and Lin, 2023), which is reported as a limitation. The date-anchored prompt does not remove what the model knows (Iacoviello and Tong, 2026), and the model's general views on places such as Iran and Hormuz may still affect scores after the cutoff (Glasserman and Lin, 2023). Other limitations are stated up front: GDELT and the AI-GPR newspapers differ as sources; crisis periods give few weekly observations, so tests there have low power; JWC listings change rarely; and Baumeister and Kilian (2016) give reason to expect supply-related signals to matter mainly when geopolitical risk coincides with tight markets, rather than on average.

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AI Declaration

Generative AI (Anthropic's Claude) was used in preparing this submission. It was used to summarise some of the papers which were too long and would take a long time to read , it was used to check the grammar of the writings in this paper and structure the paragraphs for better presentation , it was used to identify relevant literature for this project and finally the Gantt chart was also produced with the help of Generative AI.

Part 3: Critical Comparison with ChatGPT

3.1 Prompt

ChatGPT , web search enabled, 27 October 2026

"Write a literature survey on forecasting the direction of Brent crude oil prices using machine learning, covering: (1) what drives oil prices and whether they can be forecast, (2) how geopolitical risk is measured from news text using large language models, and (3) tree-based ensemble methods (Random Forest, XGBoost, LightGBM) and time-series evaluation methods for this kind of forecasting. Include citations to the academic papers it draws on."

3.2 Output

Literature Survey: Forecasting the Direction of Brent Crude Oil Prices Using Machine Learning and News-Based Geopolitical Risk

Abstract

Forecasting the direction of Brent crude oil prices is challenging because oil markets respond to supply and demand fundamentals, macroeconomic conditions, financial market dynamics and geopolitical events. Although traditional econometric models capture important features of oil price behaviour, machine learning methods can identify nonlinear relationships and interactions between a large number of predictors. Recent research also suggests that textual information from financial and energy news can improve oil futures forecasting, particularly when combined with market data.

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This literature survey examines three interconnected areas: the economic determinants and predictability of oil prices; the measurement of geopolitical risk using news text and large language models (LLMs); and the application of Random Forest, XGBoost and LightGBM to directional forecasting. It also reviews appropriate time-series validation and evaluation methods. The central argument is that combining market fundamentals with a carefully constructed, time-aligned geopolitical risk indicator may improve the prediction of Brent price direction. However, predictive improvements must be demonstrated through rigorous out-of-sample testing, rather than assumed from model complexity or in-sample accuracy.

1. Introduction

Brent crude oil is an important international benchmark whose price influences energy costs, inflation, corporate profitability and financial markets. Its movements reflect both expected physical market conditions and changes in investors' perceptions of future supply, demand and risk. Unlike many conventional forecasting problems, oil price forecasting must account for sudden structural changes, including wars, sanctions, production decisions and disruptions to transportation infrastructure.

A useful distinction exists between forecasting the price level, forecasting returns and forecasting direction. Price-level forecasting estimates the future price in monetary terms,

while return forecasting estimates the magnitude of a price change. Directional forecasting instead predicts whether the price will rise or fall over a specified horizon. For example, a model may predict whether the next trading day's Brent closing price will be higher than the current closing price.

Directional forecasting is particularly relevant when the objective is to identify short-term market movements rather than estimate an exact future price. It also introduces specific methodological challenges: a model can achieve reasonable overall accuracy simply by repeatedly predicting the more common direction, while small changes in the evaluation period can produce very different results.

This survey therefore focuses on the following research question:

To what extent can news-derived geopolitical risk, combined with conventional market indicators, improve the out-of-sample prediction of Brent crude oil price direction using tree-based ensemble methods?

The literature provides a basis for this approach, although evidence that news text improves oil price forecasting does not automatically establish that LLM-derived geopolitical risk will improve directional accuracy.

2. What drives oil prices, and can they be forecast?

2.1 Fundamental determinants of crude oil prices

A central contribution to the oil economics literature is Kilian's (2009) argument that oil price movements should not be attributed to a single type of shock. Instead, oil prices respond to three broad categories of disturbances: oil supply shocks, aggregate demand shocks and oil-market-specific demand shocks.



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Oil supply shocks include unexpected changes in production, OPEC+ production decisions, sanctions, infrastructure damage and disruptions to shipping routes. When these events reduce expected available supply, they can push prices higher, particularly when spare production capacity and inventories are limited. However, geopolitical events do not always increase prices: the market response depends on whether a disruption actually occurs, its expected duration and whether alternative suppliers can compensate.

Aggregate demand shocks arise from changes in global economic activity. Stronger industrial production and transport demand can raise oil consumption and prices, whereas economic slowdowns can reduce expected consumption. Kilian (2009) demonstrates why distinguishing demand-driven and supply-driven oil price changes is essential when interpreting market movements.



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Oil-market-specific demand shocks reflect changes in demand for oil inventories and shifts in expectations about future scarcity. Because crude oil is both a physical commodity and a

storable asset, expectations can influence current prices even before a physical supply shortage emerges.

Other relevant variables include:

- Inventories and futures-market structure: inventory levels, futures spreads and the relationship between spot and futures prices provide information about current availability and expected scarcity.
- Macroeconomic and financial conditions: US dollar movements, interest rates, inflation expectations, equity-market conditions and risk appetite can affect commodity demand and investment behaviour.
- Policy and producer decisions: OPEC+ announcements, sanctions, strategic petroleum reserve releases and changes in production capacity can alter expected supply.
- Geopolitical risk: wars, threats to energy infrastructure, sanctions and restrictions on maritime trade can change the perceived probability of future supply disruptions.

These drivers interact. For example, geopolitical tensions may increase the probability of supply disruption, but a simultaneous deterioration in global growth may offset the resulting upward pressure on prices.

2.2 Why oil price forecasting remains difficult

Baumeister and Kilian (2016), in Forty Years of Oil Price Fluctuations: Why the Price of Oil May Still Surprise Us, explain that oil price shocks remain difficult to predict even when their economic causes are better understood. The distinction between an identifiable economic mechanism and a predictable future price movement is fundamental.



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Their analysis of the 2014 oil price decline also demonstrates that some price movements were predictable from earlier supply and demand developments, while other movements reflected unexpected changes in expectations and economic conditions.



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This distinction has important implications for machine learning. Historical data may reveal recurring patterns, but a model cannot reliably anticipate genuinely unforeseen events merely by learning from previous observations. Structural changes can also weaken relationships that appeared stable in the training sample.

Nevertheless, forecasting is not necessarily futile. Costa et al. (2021), in Machine Learning and Oil Price Point and Density Forecasting, examine multiple machine learning techniques using a broad set of economic and financial variables. Their results indicate that machine learning can improve on conventional benchmarks in some forecasting settings, with reported out-of-sample R^2 values reaching 42% at a one-month horizon for certain specifications. These findings demonstrate potential predictability, rather than a universal ability to forecast oil prices accurately.

The practical implication is that the research objective should not be to establish that oil prices are entirely predictable. Instead, it should test whether a specific information set contains incremental predictive information about future Brent price direction relative to simple, defensible benchmarks.

2.3 Why directional forecasting is appropriate

For a forecasting horizon of h trading days, a binary target can be defined as:

$$ y_{t,h}= \begin{cases} 1, & P_{t+h}>P_t\\ 0, & P_{t+h}\leq P_t \end{cases} $$

where P_t denotes the Brent price observed at time t.

This formulation transforms the problem into binary classification. A model estimates the probability of an upward move and predicts a direction using a threshold, commonly 0.5.

Directional forecasting has several advantages. It aligns directly with the question of whether the market will rise or fall, avoids requiring precise forecasts of price magnitude and allows performance to be evaluated using classification metrics.

However, it discards information about the size of price changes. Predicting a one-cent increase and a substantial price surge correctly receives the same directional label. It is therefore useful to supplement directional classification with return-based or economic evaluations where appropriate.

3. Measuring geopolitical risk from news text using large language models

3.1 Traditional news-based geopolitical risk measurement

A foundational contribution is Caldara and Iacoviello's (2022) Measuring Geopolitical Risk, published in the American Economic Review. The authors develop a news-based geopolitical risk (GPR) index by measuring the frequency of newspaper articles containing terms associated with geopolitical tensions and adverse geopolitical events. The resulting index rises around major historical episodes, including the world wars, the Cuban Missile Crisis and the September 2001 attacks.

The GPR index provides a systematic way to quantify geopolitical risk over time, avoiding exclusive reliance on retrospective descriptions of individual events. However, a frequency-based index has limitations. The number of articles mentioning geopolitical risk does not necessarily reveal whether an event threatens oil supply, whether the threat is credible or whether the information is genuinely new to financial markets.

A similar principle appears in Baker, Bloom and Davis's (2016) economic policy uncertainty index, which uses newspaper coverage to quantify uncertainty related to economic policy. Their work illustrates how textual information can be converted into a quantitative measure that can subsequently be incorporated into economic models.

For oil price forecasting, the distinction between general geopolitical risk and oil-relevant geopolitical risk is important. A geopolitical event may be highly significant internationally without materially affecting crude oil production, consumption or transportation.

3.2 From keyword counts to LLM-based risk extraction

Large language models offer a potential extension to traditional news-based indices because they can interpret context, relationships and the meaning of an event rather than simply count matching words.

For example, consider two hypothetical headlines:

- “Diplomatic tensions increase between two countries.”
- “New sanctions threaten crude oil exports from a major producer.”

Both may contain geopolitical language, but the second headline more directly indicates a potential oil supply shock. An LLM-based method can be designed to distinguish between them and extract information about the mechanism through which an event could affect Brent prices.

A suitable workflow comprises the following stages.

Stage 1 — News collection

Collect timestamped news articles from reliable sources, including headlines, publication times and article text where available.

Stage 2 — Relevance filtering

Identify articles concerning conflicts, sanctions, production facilities, pipelines, shipping routes, OPEC+ and other developments relevant to oil supply or demand.

Stage 3 — LLM-based classification

Classify each article by geopolitical relevance, oil-market relevance, event type, expected supply or demand effect, and the direction and severity of the potential impact.

Stage 4 — Daily risk aggregation

Convert article-level classifications into daily indicators, such as average risk scores, counts of high-severity events and separate threat and realised-event measures.

Stage 5 — Forecasting integration

Combine the news-derived indicators with lagged Brent returns and other market variables to predict the subsequent direction of Brent prices.

This pipeline is a proposed research design, not a claim that LLMs have already been shown to outperform the traditional GPR index for Brent directional forecasting.

3.3 Designing an LLM-based geopolitical risk indicator

The classification prompt should define the task explicitly and require a structured output. One possible schema is:

Feature	Description
geo_relevance	Whether the article concerns geopolitical developments
oil_relevance	Degree of relevance to the global oil market
event_type	Conflict, sanctions, production disruption, shipping, diplomacy or other
risk_severity	Ordinal severity score, such as 0–4
supply_impact	Expected effect on oil supply: negative, neutral or positive
demand_impact	Expected effect on oil demand: negative, neutral or positive
event_status	Threat, realised disruption or de-escalation
confidence	Model's stated confidence in the classification

Supply and demand effects should be recorded separately because they can point in different directions. A conflict could threaten supply while simultaneously weakening expected global demand.

The daily index can then be constructed from article-level scores. For example:

\[GPR_t=\frac{\sum_{i=1}^{N_t} w_i s_i} {\sum_{i=1}^{N_t} w_i} \]

where \(s_i\) is the risk score assigned to article \(i\), \(w_i\) is an optional relevance or source-quality weight and \(N_t\) is the number of eligible articles published on day \(t\).

A weighted average measures the average severity of reported risk, while a weighted sum may better capture the total volume of risk-related coverage. Both should be considered because article volume and event severity represent different dimensions. If no eligible articles are published, the index should distinguish missing news coverage from an observed absence of risk.

For greater robustness, the index could also include separate measures of oil-supply threats, realised disruptions and de-escalation. This would allow the model to learn whether different types of geopolitical news have distinct predictive relationships with Brent prices.

3.4 What does the existing literature say about text-based oil forecasting?

Research directly supports the proposition that textual information can improve oil-market forecasting.

Gong, Guan and Chen (2022), in The Role of Textual Analysis in Oil Futures Price Forecasting Based on Machine Learning Approach, find that text-derived features from online oil news complement conventional predictors and improve forecasting performance. Their event-study results also indicate asymmetric responses to positive and negative news shocks.


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Pan, Zhong, Wang and Huang (2024), in Forecasting Oil Futures Returns with News, develop a mixed-frequency data sampling Random Forest approach to extract information

from news. They report improved forecasting performance from their textual indicator, including a monthly out-of-sample \(R^2\) of 5.26% and an annualised certainty-equivalent return gain of 3.08% in their evaluation setting. Their analysis suggests that capital-market and macroeconomic topics contribute substantially to the indicator's predictive value.



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These studies provide an empirical foundation for incorporating news into an oil forecasting system. However, they do not establish that geopolitical risk alone is responsible for the improvement, nor do they directly validate an LLM-derived index for Brent directional classification.

An LLM-based indicator should therefore be evaluated against at least three alternatives:

1. A traditional keyword-frequency geopolitical risk measure.
2. A conventional sentiment or news-topic indicator.
3. The proposed LLM-based, oil-specific geopolitical risk indicator.

The third approach would be most useful if it provides measurable incremental predictive value beyond the first two.

3.5 Limitations of LLM-based measurement

Several issues require careful treatment.

Context and classification errors. LLMs can confuse hypothetical threats with realised disruptions, misinterpret negation or assign excessive severity to dramatic headlines. Human-labelled validation data are therefore necessary.

News duplication. Multiple publications may report the same event. Counting every article independently can inflate the apparent level of risk. Near-duplicate detection and event-level aggregation can reduce this problem.

Look-ahead bias. An article's publication timestamp must determine when its information becomes available. A model must not use an article published after the forecast cutoff, even if its date matches the day being forecast.

Model and prompt stability. Changes in model versions, prompts or output behaviour can produce changes in the index unrelated to actual geopolitical conditions. The model, prompt, scoring rules and aggregation method should be fixed and documented during evaluation.

Validation and reproducibility. A manually labelled sample should be used to assess precision, recall and agreement between the LLM and human annotators. The index should also be tested for stability across alternative prompts and, where feasible, independent models.

Taken together, the literature supports the use of news text as an additional information source, while the specific advantage of LLM-based geopolitical risk extraction remains an empirical question.

4. Tree-based ensemble methods for directional forecasting

Tree-based ensemble methods are attractive for Brent price forecasting because they can capture nonlinear relationships, interactions between predictors and threshold effects without requiring the researcher to specify a complete parametric model.

For example, the effect of geopolitical risk may depend on the level of oil inventories, the prevailing market trend or the degree of financial uncertainty. A tree-based model can learn these conditional relationships from historical observations.

4.1 Random Forest

Breiman (2001) introduced Random Forests as an ensemble of decision trees constructed using randomised samples and subsets of features. The predictions of individual trees are aggregated to produce a final estimate.



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For directional forecasting, Random Forest can be used as a classifier. Each tree predicts whether Brent will rise or fall, and the model combines the predictions into an estimated probability of an upward move.

Its principal advantages include:

- Nonlinear relationships: it can identify thresholds and interactions between market and news indicators.
- Resistance to overfitting: averaging predictions across diverse trees generally reduces the variance associated with individual decision trees.
- Limited preprocessing: it does not require predictors to be normally distributed or standardised.
- Interpretability tools: feature importance and SHAP values can help identify which variables contribute to predictions.

However, Random Forest does not automatically model temporal dependence. Lagged returns, volatility, futures spreads and news-derived indicators must be supplied as predictors. Its performance may also deteriorate when market relationships change sharply, and conventional feature importance can be misleading when predictors are strongly correlated.

Evidence from oil forecasting is encouraging. Costa et al. (2021) include Random Forest among the methods examined for point and density forecasting, while Pan et al. (2024) develop a Random Forest framework that incorporates mixed-frequency news information.



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4.2 XGBoost

XGBoost, introduced by Chen and Guestrin (2016), is a scalable implementation of gradient-boosted decision trees. Rather than independently constructing trees and averaging their outputs, boosting builds trees sequentially, with subsequent trees improving

the model's existing predictions. Its objective function incorporates regularisation to control model complexity.



DOI

For Brent directional classification, XGBoost can use a logistic classification objective to estimate:

\[\hat p_t=\Pr(y_{t,h}=1\mid X_t) \]

where \(X_t\) contains the information available at the forekast time.

XGBoost is particularly useful when the predictor set combines continuous market variables, binary event indicators, lagged features and aggregated text scores. Its regularisation parameters, tree depth, learning rate and subsampling options provide several ways to control overfitting.

Nevertheless, a large number of hyperparameters creates a risk of excessive tuning to historical market conditions. Hyperparameter selection must therefore be performed using chronological validation data rather than the final test set.

4.3 LightGBM

Ke et al. (2017) introduced LightGBM as an efficient gradient-boosting framework. Its principal innovations include Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB), which reduce computational costs when training on large datasets or high-dimensional features.



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LightGBM is a natural candidate for a news-enhanced oil forecasting system because textual features can generate a relatively large predictor set, particularly when the research includes multiple risk categories, sentiment measures, lags and rolling statistics.

Its advantages include efficient training and the ability to learn complex interactions. However, its leaf-wise tree growth can produce overly complex trees on small datasets unless parameters such as the number of leaves, minimum observations per leaf and regularisation are appropriately constrained.

Gong et al. (2022) provide direct evidence for the relevance of LightGBM to oil futures forecasting. Their study finds that text-derived news features improve forecasting performance when incorporated into LightGBM and benchmark models.



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4.4 Comparing the three methods

Characteristic	Random Forest	XGBoost	LightGBM

Ensemble strategy	Bagging and averaging	Sequential boosting	Sequential boosting
Main strength	Robust baseline; relatively straightforward	Flexible, regularised learning	Efficient learning with many features
Nonlinear interactions	Yes	Yes	Yes
Tuning sensitivity	Moderate	Potentially high	Potentially high
Main concern	May not capture subtle patterns efficiently	Overfitting through excessive tuning	Overfitting with overly complex leaf structures
Role in this study	Robust benchmark	Main boosted-tree comparator	Main efficiency-oriented boosted-tree comparator

No method should be assumed to perform best in advance. The optimal model depends on the data frequency, forecasting horizon, feature construction and degree of structural change in the market.

A particularly useful design is to compare all three methods on the same features, target labels, training periods and test observations. This isolates differences in learning algorithms from differences in data preparation.

5. Time-series evaluation and experimental design

Forecast evaluation is central to the credibility of this research. A high classification score is not persuasive if the model has inadvertently used future information or has been tuned to the test period.

5.1 Chronological train, validation and test sets

Unlike an ordinary independent-data classification problem, oil price forecasting must preserve the order of observations.

A suitable design divides the historical dataset into three consecutive periods:

1. Training period: fit the model and estimate its parameters.
2. Validation period: select hyperparameters and decision thresholds.
3. Test period: evaluate final out-of-sample performance without further tuning.

The final test period must remain untouched until model selection is complete. The precise dates and proportions should reflect the available sample size and the need to include different market conditions.

5.2 Rolling-origin evaluation

Hyndman and Athanasopoulos describe time-series cross-validation as an approach in which each forecast is generated using only observations available before the forecast origin. The process is repeated as the origin moves forward in time.


Forecasting: Principles and Practice (3rd ed)+1

Two common configurations are:

- Expanding window: retain all historical training observations and add new observations as time advances.
- Rolling window: retain a fixed-length training period and discard older observations as the window moves.

An expanding window uses more historical information, while a rolling window can adapt more readily to changing market conditions. Both are defensible and can be compared as a robustness check.

Bergmeir, Hyndman and Koo (2018) show that ordinary K-fold cross-validation can be valid under particular autoregressive modelling conditions. Nevertheless, for this study, chronological rolling-origin evaluation is the clearer primary design because it directly represents the information available to a real-time forecaster.


Rob J Hyndman

5.3 Directional classification metrics

Let \(TP\), \(TN\), \(FP\) and \(FN\) represent true positives, true negatives, false positives and false negatives, respectively.

The following metrics provide complementary perspectives:

Metric	Definition	Interpretation
Accuracy	\((TP+TN)/N\)	Overall proportion of correct predictions
Precision	\(TP/(TP+FP)\)	Proportion of predicted upward moves that actually rise
Recall	\(TP/(TP+FN)\)	Proportion of actual upward moves correctly identified
F1-score	\(2PR/(P+R)\)	Balance between precision and recall
ROC-AUC	Area under ROC curve	Ability to rank upward moves above downward moves
Balanced accuracy	Mean of the two class recalls	Performance adjusted for class imbalance
Log loss	Average negative log-likelihood	Quality of the predicted probabilities

Accuracy should not be interpreted in isolation. If upward moves occur in 55% of observations, a model that always predicts an increase achieves 55% accuracy without identifying any useful market signal. The majority-class predictor and a simple historical-direction benchmark should therefore be included.

ROC-AUC is useful because it evaluates ranking performance across thresholds, but it does not establish that predicted probabilities are well calibrated. Log loss and, where appropriate, calibration plots provide additional information about probability quality.

5.4 Comparing predictive performance statistically

Differences in observed accuracy can arise by chance, particularly when the test sample is small or market conditions change considerably.

Diebold and Mariano (1995) develop a statistical framework for testing whether two competing forecasts have different expected predictive accuracy. The test compares a loss differential between forecasts and can accommodate several loss functions.


DOI

For this research, the comparison could assess whether the LLM-enhanced LightGBM model performs better than a market-only LightGBM model. Depending on the forecast target and loss function, a suitable comparison may use classification losses or probabilistic losses such as log loss.

However, overlapping multi-day forecast horizons can generate serial dependence in forecast errors. Statistical inference must account for that dependence, and multiple model comparisons should be interpreted cautiously.

5.5 Avoiding look-ahead bias in news-based forecasting

News data create a particular risk of leakage. A daily article published after the market closes cannot legitimately contribute to a prediction supposedly made before that day's close.

A reliable implementation should:

- Assign every article a publication timestamp and a clearly defined market-information cutoff.
- Aggregate only information available at or before the forecast origin.
- Use lagged market indicators when their latest observations would not yet have been available.
- Fit scaling, imputation and feature-selection procedures using training data only.
- Tune hyperparameters without consulting the final test set.
- Avoid using revised or retrospectively constructed geopolitical indices unless their historical availability is verified.

These controls are essential because even a modest amount of look-ahead bias can create apparently strong results in financial forecasting.

6. Proposed empirical framework and research gap

The literature suggests that a useful contribution would be to test whether an LLM-derived geopolitical risk indicator improves Brent directional forecasting beyond information already contained in market prices and conventional risk indicators.

6.1 Proposed model design

The experimental design should compare models with progressively richer information sets.

Model	Predictors	Purpose
M0: Naive benchmark	Majority-class and historical-direction benchmarks	Establish minimum performance

M1: Market-only	Lagged Brent returns, volatility, futures spreads and selected macro-financial variables	Measure predictive value of market information
M2: Traditional risk	M1 plus the established GPR index	Test whether conventional geopolitical risk adds value
M3: News sentiment	M1 plus traditional news sentiment and topic indicators	Test whether broader news information adds value
M4: LLM geopolitical risk	M1 plus LLM-derived oil-specific geopolitical risk	Test the proposed contribution
M5: Full model	M1 plus conventional GPR, news sentiment and LLM-derived risk	Test whether the information sources are complementary

Each model should be evaluated using Random Forest, XGBoost and LightGBM, with hyperparameters selected through chronological validation.

The key comparison is not simply whether M4 achieves high accuracy. It is whether M4 or M5 consistently outperforms M1 and M2 on genuinely unseen data.

6.2 Feature selection and forecast horizon

The market feature set should be deliberately limited to variables with a clear economic justification. Possible predictors include:

- Lagged Brent returns over several trading days.
- Rolling volatility and recent price momentum.
- Front-month futures spreads or other indicators of market structure.
- Relevant macro-financial variables, such as the US dollar and financial market volatility.
- The traditional GPR index.
- Daily LLM-derived geopolitical risk, oil-supply threat and de-escalation indicators.

The forecast horizon should be fixed before the final evaluation. A one-day-ahead forecast is suitable for testing immediate market reactions, while a five-day horizon can assess whether geopolitical information has a more persistent relationship with price direction. Results across horizons should be reported separately.

6.3 Interpreting model decisions

Feature importance and SHAP analysis can help determine whether geopolitical risk contributes meaningfully to model predictions.

For example, the analysis could investigate whether the LLM-derived risk score becomes more influential during periods of elevated market volatility, or whether supply-disruption indicators are associated with upward predictions more strongly when inventories are low.

Such findings would help connect statistical performance with plausible economic mechanisms. Nevertheless, feature importance and SHAP values describe model behaviour, not causal effects. They cannot independently establish that geopolitical risk caused a subsequent price change.

6.4 The research gap

The reviewed literature establishes three relevant findings.

First, oil prices reflect multiple interacting supply, demand and expectation shocks, making their movements difficult but not necessarily impossible to forecast (Kilian, 2009; Baumeister and Kilian, 2016).

Second, news-derived indicators can improve oil futures forecasting in certain settings (Gong et al., 2022; Pan et al., 2024). Research on financial text also suggests that language models can capture contextual sentiment more effectively than simple word-count approaches in some tasks (Huang et al., 2023; Kirtac and Germano, 2024).


American Economic Association+5

Third, Random Forest, XGBoost and LightGBM provide complementary approaches for modelling nonlinear relationships and interactions between market and textual features.

The remaining question is whether LLM-derived, oil-specific geopolitical risk provides incremental out-of-sample predictive value for Brent price direction beyond conventional market variables and established news-based risk measures.

This is a focused and testable research gap. The contribution would not simply be to apply a new machine learning algorithm, but to evaluate a new source of information under rigorous time-series conditions.

7. Conclusion

The literature supports a combined approach to forecasting Brent crude oil direction, using conventional market information alongside news-derived geopolitical risk. Oil prices respond to supply disruptions, global demand, inventories, expectations and financial conditions, while geopolitical developments can alter the perceived probability of future disruptions before their physical effects materialise.

Traditional news-based indices provide a useful benchmark, but LLMs may offer a more contextual way to distinguish general geopolitical coverage from events that are specifically relevant to oil supply and demand. Existing research on news-based oil forecasting supports the value of textual features, although the additional predictive value of LLM-derived geopolitical risk for Brent directional classification remains to be established.

Random Forest, XGBoost and LightGBM are appropriate candidate models because they can learn nonlinear interactions without requiring a fully specified economic model. Their performance should be compared using chronological validation, a held-out test period, directional classification metrics and statistical tests of predictive differences.

Ultimately, the strongest evidence would be a model that consistently outperforms market-only and conventional-risk benchmarks across multiple test periods, while remaining robust to changes in market conditions. Such a result would support the practical value of LLM-based geopolitical risk measurement; failure to improve on the benchmarks would also be an informative finding about the limits of extracting additional predictive information from news.

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3.3 Observations

The output is well organised and reads like a competent survey and its citations are more reliable than had expected. ChatGPT searched the web while answering and all sixteen references point to real papers. Where checked, the details match the sources: the figures attributed to Pan et al. (2024), including an out of sample R^2 of 5.26%, match the paper’s abstract and Gong, Guan and Chen (2022) do use LightGBM with text features as described. One figure, a 42% out-of-sample R^2 for Costa et al. (2021) could not be confirmed from the abstract. The general methodological advice is also good: chronological validation, a majority-class baseline, careful timestamping of news and SHAP values treated as descriptions of the model rather than causal effects.

The weaknesses are in the content. Kilian (2009) is misrepresented by omission. The survey describes sanctions and shipping disruptions as supply shocks that raise prices but never states his main finding that supply shocks had small and transitory effects with most large price rises driven by demand and precautionary demand. Its section on look ahead bias covers only article timestamps and misses the problem specific to LLMs: a model scoring news from before its training cutoff may already know what happened next (Glasserman and Lin, 2023). Most importantly, it proposes an LLM derived, oil specific geopolitical risk index as an open research gap without mentioning the AI-GPR index (Iacoviello and Tong, 2026) which already provides one or the other 2026 studies closest to this project. Its evaluation section also omits small sample corrections to the Diebold–Mariano test. Overall, the output is a useful starting outline but no substitute for reading the papers since its errors are omissions that only someone who knows the literature would notice.